

Chapter 05: Elementary functions

1- Inverse trigonometric functions:

1-1- $\text{Arccos}(\cos^{-1})$:

The function $f: [0; \pi] \longrightarrow [-1; 1]$
 $x \longrightarrow \cos x$

f is $\begin{cases} \text{monotone} \\ \text{continuous} \end{cases}$ function

by Monotone bijective Theorem:

The Inverse funct $f: [-1; 1] \longrightarrow [0; \pi]$
 $x \longrightarrow \text{Arccos } x$

f^{-1} is $\begin{cases} \text{Monotone} \\ \text{continuous} \end{cases}$ function

properties:

- ① $\cos(\text{Arccos } x) = x ; \forall x \in [-1; 1]$
- ② $\text{Arccos}(\cos x) = x ; \forall x \in [0; \pi]$
- ③ $(\text{Arccos } x)' = \frac{-1}{\sqrt{1-x^2}} ; \forall x \in]-1; 1[$

Derivative of $\text{Arc cos}(u(x))$:

$$[\text{Arccos}(u(x))]' = \frac{-u'(x)}{\sqrt{1-u^2(x)}}$$

proof:

we know that: $f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$

we have: $\rightarrow f(x) = \cos x$
 $\rightarrow f'(x) = -\sin x$

$$[f^{-1}(x)]' = (\text{Arccos } x)'$$

$$= \frac{-1}{\sin(\text{Arccos } x)}$$

$$(\text{Arccos } x)' = \frac{-1}{\sin(\cos^{-1} x)} = \frac{-1}{\sqrt{1-x^2}}$$

$$\cos^2 x + \sin^2 x = 1$$

$$\sin^2 x = 1 - \cos^2 x \rightarrow \sin x = \sqrt{1 - \cos^2 x}$$

$$\sin(\cos^{-1} x) = \sqrt{1 - (\cos(\cos^{-1} x))^2}$$

$$= \sqrt{1 - x^2} / u(x), x \text{ نفس الشيء بس } x$$

2-2- $\text{Arc sin}(\sin^{-1})$:

The function $f: [-\frac{\pi}{2}; \frac{\pi}{2}] \longrightarrow [-1; 1]$
 $x \longrightarrow \sin x$

f is $\begin{cases} \text{monotone} \\ \text{continuous} \end{cases}$ function

by Monotone bijective Theorem:

The Inverse funct $f: [-1; 1] \longrightarrow [-\frac{\pi}{2}; \frac{\pi}{2}]$
 $x \longrightarrow \text{Arc sin } x$

f^{-1} is $\begin{cases} \text{Monotone} \\ \text{continuous} \end{cases}$ function

properties:

- ① $\sin(\text{Arc sin } x) = x ; \forall x \in [-1; 1]$
- ② $\text{Arc sin}(\sin x) = x ; \forall x \in [-\frac{\pi}{2}; \frac{\pi}{2}]$
- ③ $(\text{Arc sin})' = \frac{1}{\sqrt{1-x^2}}$

Derivative of $\text{Arc sin } u(x)$:

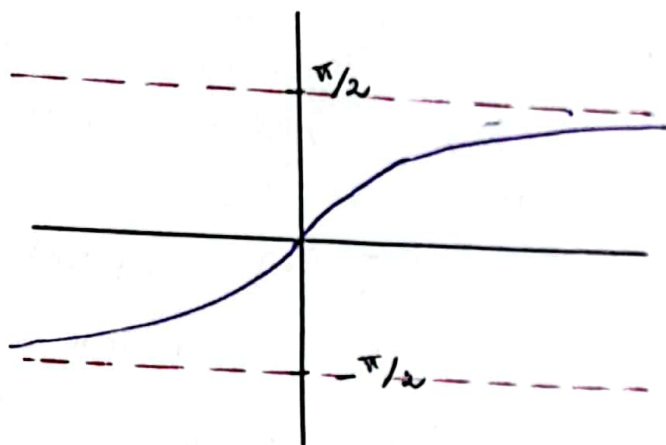
$$(\text{Arc sin } u(x))' = \frac{u(x)}{\sqrt{1-u(x)^2}}$$

1.3 - $\text{Arctan } (\tan^{-1})$:

$$f = \tan x, f:]-\frac{\pi}{2}; \frac{\pi}{2}[\rightarrow \mathbb{R}$$

$$f^{-1}: \text{Arctan} = f^{-1}: \mathbb{R} \rightarrow]-\frac{\pi}{2}; \frac{\pi}{2}[$$

$$x \rightarrow \text{Arctan } x$$



properties:

$$\textcircled{1} \tan(\text{Arctan } x) = x, \forall x \in \mathbb{R}$$

$$\textcircled{2} \text{Arctan}(\tan x) = x, \forall x \in]-\frac{\pi}{2}; \frac{\pi}{2}[$$

$$\textcircled{3} [\text{Arctan}(x)]' = \frac{1}{x^2+1}$$

Derivative of $\text{Arc tan } u(x)$:

$$[\text{Arctan } u(x)]' = \frac{1}{[u(x)]^2+1}$$

Note:

$$\text{Arc tan } x + \text{Arc tan } \frac{1}{x} = \frac{\pi}{2}$$

2 - Hyperbolic functions:

Def:

① Hyperbolic cosine x :

$$\cosh x = \frac{e^x + e^{-x}}{2} = \text{Ch } x$$

② Hyperbolic sine x :

$$\sinh x = \frac{e^x - e^{-x}}{2} = \text{Sh } x$$

③ Hyperbolic tangent x :

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{\text{Sh } x}{\text{Ch } x} = \text{th } x$$

Propositions:

$$\textcircled{1} \cosh(-x) = \cosh(x)$$

$$\textcircled{2} \sinh(-x) = -\sinh(x)$$

$$\textcircled{3} \rightarrow \cosh(0) = 1$$

$$\rightarrow \sinh(0) = 0$$

$$\textcircled{4} \rightarrow [\cosh(x)]' = \sinh(x)$$

$$\rightarrow [\sinh(x)]' = \cosh(x)$$

Theorem:

$$\forall x \in \mathbb{R}; \cosh^2 x - \sinh^2 x = 1$$

2.1 - Argch :

$$f:]0; +\infty[\rightarrow]1; +\infty[$$

$$x \rightarrow \cosh x$$

$$f^{-1}:]1; +\infty[\rightarrow]0; +\infty[$$

$$x \rightarrow \text{Argch } x (\cosh^{-1})$$

$$\hookrightarrow [\cosh(x)]^{-1} = \text{Argch } x$$

$$[\cosh(x)]^{-1} = \text{Argch } x$$

Derivatives

$$[\operatorname{Argch}(u(x))]' = \frac{u'(x)}{\sqrt{u^2(x) - 1}}$$

2-2 - Argsh:

$$f: \begin{array}{ccc} \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longrightarrow & \operatorname{sh} x \end{array}$$

$$f^{-1}: \begin{array}{ccc} \mathbb{R} & \longrightarrow & \mathbb{R} \\ x & \longrightarrow & \operatorname{Argsh} x (\operatorname{sh} x^{-1}) \end{array}$$

Derivative:

$$[\operatorname{Argsh}(u(x))]' = \frac{u'(x)}{\sqrt{u^2(x) + 1}}$$

2-3 - Arcth:

$$f: \begin{array}{ccc} \mathbb{R} & \longrightarrow &]-1; 1[\\ x & \longrightarrow & \tanh(x) \end{array}$$

$$f^{-1}: \begin{array}{ccc}]-1; 1[& \longrightarrow & \mathbb{R} \\ x & \longrightarrow & \operatorname{Arctanh}(x) \end{array}$$

Derivative:

$$[\operatorname{Arctanh}(u(x))]' = \frac{1}{1 - [u(x)]^2}$$